

Budget-Friendly Oscilloscopes

Digital oscilloscopes, being expensive, have always been out of reach for many. But this doesn't make them less desirable. Fortunately, test and measurement companies have sensed this situation and started developing low-priced oscilloscopes with reasonable quality and features. So we now have several wonderful low-budget options. Let us take a look at some of them



ANKIT GUPTA

A good oscilloscope is an indispensable tool for you if you are seriously into electronics. The oscilloscope helps you check the functioning of electronic circuits better. With advancements in technology, electronics designing, testing and even repairing has become much more challenging than before. Engineers thus require advanced tools to solve measurement challenges quickly and accurately, and a digital storage oscilloscope helps them do just that. But, due to its high price, this equipment has always been out of reach for most individuals.

The strong desire of the engineers to possess a good oscilloscope has led companies such as Tektronix, Agilent

and Teledyne LeCroy to introduce low-priced oscilloscopes that promise reasonable acquisition quality and performance. Generally, these look like a standard oscilloscope with integrated display for viewing the acquired signal as shown in Fig. 1.

Another low-cost solution comes from companies such as Digilent, Virtins Technology, Link Instruments, Parallax Inc. and Velleman Inc., which uses processing power of a computer



Fig. 1: An oscilloscope with integrated display

and its monitor to display the acquired signal. These solutions are becoming more popular as they are compact yet show the signal on a larger display of a computer for better analysis. Some of these are even multifunctional, with additional functions of waveform generator, logic analyser, DC supply and multimeter. Fig. 2 shows such a system.

Next are the small pocket-size oscilloscopes that can be highly useful for field engineers. Most of these are low-priced. Though their accuracy is not as good as the previous options but these are good enough for basic analysis. Fig. 3 shows a pocket-size oscilloscope.

There are also various open source do it yourself (DIY) oscilloscope designs available on the Internet. You can build these systems yourself and use for basic test and measurement. Such systems are very useful for students as they learn a lot while building their own oscilloscope and then use it with better confidence. Fig. 4 shows such a system.

There are so many options avail-



Fig. 2: An oscilloscope that shows signal on a computer's screen